GLA University, Mathura
Department of Computer Engineering and Applications
Subject: Computer Organization
Subject Code: BCSC1005
Submission deadline: **November 30, 2025**

Assignment 2.1 (Module II)

1. In a certain scientific computation, it is necessary to perform the arithmetic operations $A_i * B_i * (C_i * D_i)$ with a stream of numbers. Specify a pipeline configuration to carry out this task.
2. A four-stage pipeline has the stage delays as 150, 120, 160 and 140 ns respectively. Registers are used between the stages and have a delay of 5 ns each. Assuming constant clocking rate, what will be the total time taken to process 1000 data items on the pipeline? Also calculate the speed up factor and throughput for this pipeline configuration.
3. A two-way set associated cache memory uses blocks of four words. The cache can accommodate a total of 2048 words from main memory. The main memory size is 128K × 32.
   (i) Formulate all pertinent information required to construct the cache memory (i.e. Tag, Index, Data, Block, Words)
   (ii) What is the size of cache memory.
4. A virtual memory system has an address space of 8k words, memory space of 4k words and Page & Block size of 1k words. The following page reference changes occur during a given time interval. 4, 2, 0, 1, 2, 6, 1, 4, 0, 1, 0, 2, 3, 5, 7

   Determine the four pages that are resident in main memory after each Page reference change if the replacement algorithm used is
   (i) FIFO
   (ii) LRU.
5. The access time of a cache memory is 120 ns and that of main memory 900 ns. It is estimated that 80% of the memory requests are for read and remaining 20% for write. The hit ratio for read access only is 0.9. A write-through procedure is used.
   (i) What is the average access time of the system considering only memory real cycles?
   (ii) What is the hit ratio taking in to consideration the write cycle?
   (iii) What is the average access time of the system for both read and write requests.

6. A computer uses RAM chips of 1024 x 1 capacity.

a. How many chips are needed, and how should their address lines be connected to provide a memory capacity of 1024 bytes?
b. How many chips are needed to provide a memory capacity of 16K bytes? Explain in words how the chips are to be connected to the address bus.

7. Write the sequence of micro-operations a computer system performs to execute the memory-reference instructions (i) STA (ii) ADD to AC and register-reference instructions (i) CIR (circular shift right) (ii) INC (increment AC)

8. A Virtual memory has a Page Size of 1K words. There are eight Pages and four blocks. The associative memory page table contains the following entries.

Page	Block
6	0
1	1
4	2
0	3

Make a list of all virtual addresses in decimal that will cause a Page fault if used by CPU.

9. The initial content of PC is 7FFH. The content of memory at address 7FF is 9A9FH. The content of memory at address A9F is 0C35. The content of memory address C35 is FFFF. List the contents in hexadecimal of registers Program Counter (PC), Address Register (AR), Data Register (DR), and Accumulator of the basic computer when an ADD instruction is fetched from memory and executed.

10. What are the various write policies used to write data in cache? Explain them with advantage and disadvantage of each.

11. Explain Asynchronous data transfer using Strobe & Handshaking.

12. A computer system has a 4 K word cache organized in block-set-associative manner with 4 blocks per set, 64 words per block. The number of bits in the SET and WORD fields of the main memory address format are?

13. Why are the read and write line in a DMA controller bidirectional? Under what conditions they are used as inputs and outputs? Explain with the help of suitable diagram.

14. What are the various write policies used to write data in cache? Explain them with advantage and disadvantage of each.

15. A computer engineering team is designing two different types of processors.

Processor A needs to be very fast and is intended for a real-time embedded system where performance and quick response are critical.

Processor B is designed for a general-purpose computer that must support a large and complex instruction set, and the design team wants to be able to easily update or modify the control logic in the future.

Based on this scenario, compare and explain which type of control unit — Hardwired Control Unit or Microprogrammed Control Unit — would be more suitable for each processor, and justify your answer.

16. The initial content of PC is 3BF H. the content of memory at address 3BF is 7020. List the contents of registers PC, MAR, MDR and AC (in binary) of the basic computer when an "Increment Accumulator" instruction is fetched from memory and executed. The initial content of Accumulator (AC) is DCF9 H.

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Assignment 2.2 (Module II)

1. Suppose that we want to perform the following arithmetic operations with a stream of numbers.

$(A_i + B_i)\ (C_i + D_i)$ for $i = 1, 2, 3, \ldots, 7$

Show the sub operations performed in each segment of the pipeline processing by considering different registers (R1, R2, R3, … etc.).
List the content of registers used in pipeline.
Draw a space time-diagram for a six-segment pipeline showing the time it takes to process eight tasks.

2. To execute an instruction by a 32-bit machine the following steps are carried out: Fetch, Decode, Execution, Memory access and Store, each of which takes 1 clock period. In a pipelined execution of a 5-step task, a new instruction is read and it takes 1 ns. If there are 100 instructions in sequence, then determine the speedup ratio (S) and its maximum value (Smax), of pipe line processing system over an equivalent non pipeline processing system.

3. Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given the latch delay is 10ns. Calculate the following:

(i) Pipeline cycle time and non-pipeline execution time.
(ii) Pipeline time and sequential time for 1000 tasks.
(iii) Throughput and efficiency.
(iv) Speedup ratio.

4. A four-stage pipeline has the stage delays as 150, 120, 160 and 140 ns respectively. Registers are used between the stages and have a delay of 5 ns each. Assuming constant clocking rate, calculate the total time taken to process 1000 data items on the pipeline.

5. The CPU communicates with an I/O device using asynchronous data transfer. Explain how the data transfer takes place using the strobe control and handshaking methods, illustrate

both methods with neat and labeled timing diagrams showing the sequence of control signals (strobe, acknowledge, and data lines).

6. Consider a 32-bit microprocessor that has an on-chip 16 KB four way set associative cache. Assume that the cache has a line size of four 32-bit words. Draw a block diagram of this cache showing its organization and how the different address fields are used to determine a cache hit/miss. Where in the cache is the word from memory location ABCDE8F8 mapped?

7. A computer system is designed with multiple levels of memory, including cache, main memory (RAM), and secondary storage (hard disk). During program execution, the CPU frequently accesses data and instructions stored in these different memory levels. Explain how the memory hierarchy helps in improving the overall system performance. Illustrate your answer with a neat diagram showing the memory hierarchy levels, their access time, capacity, and cost per bit. Also, describe what happens when the CPU requests data that is not found in the cache memory.

8. A microprocessor-based system is designed with a total addressable memory space of 2 KB. The system uses separate chips for RAM and ROM. The RAM section requires 512 bytes of memory, implemented using four RAM chips of equal size, while the ROM section also requires 512 bytes of memory, implemented using a single ROM chip. Assuming the memory mapping starts from address 0000H for RAM, and ROM is placed immediately after the RAM section, draw the memory map of the system and determine the address range (in hexadecimal) assigned to each RAM chip and to the ROM chip.

9. A digital computer has memory unit of 64K * 16 and cache memory of 1K words. The cache uses direct mapping with a block size of four words.

(i) How many bits are there in the tag, index, block & word fields of the address formula?

(ii) How many bits are there in each word of cache & how are they divided into functions? Include a valid bit.

(iii)How many blocks can be the accommodate?

10. A two - way set associative cache memory uses blocks of four words. The cache can accommodate a total of 2048 words from main memory. The main memory size is 128K*32.

(i) Formulate all information required to construct the cache memory. (Tag, index, data, block, words).

(ii) What is the size of the cache memory.

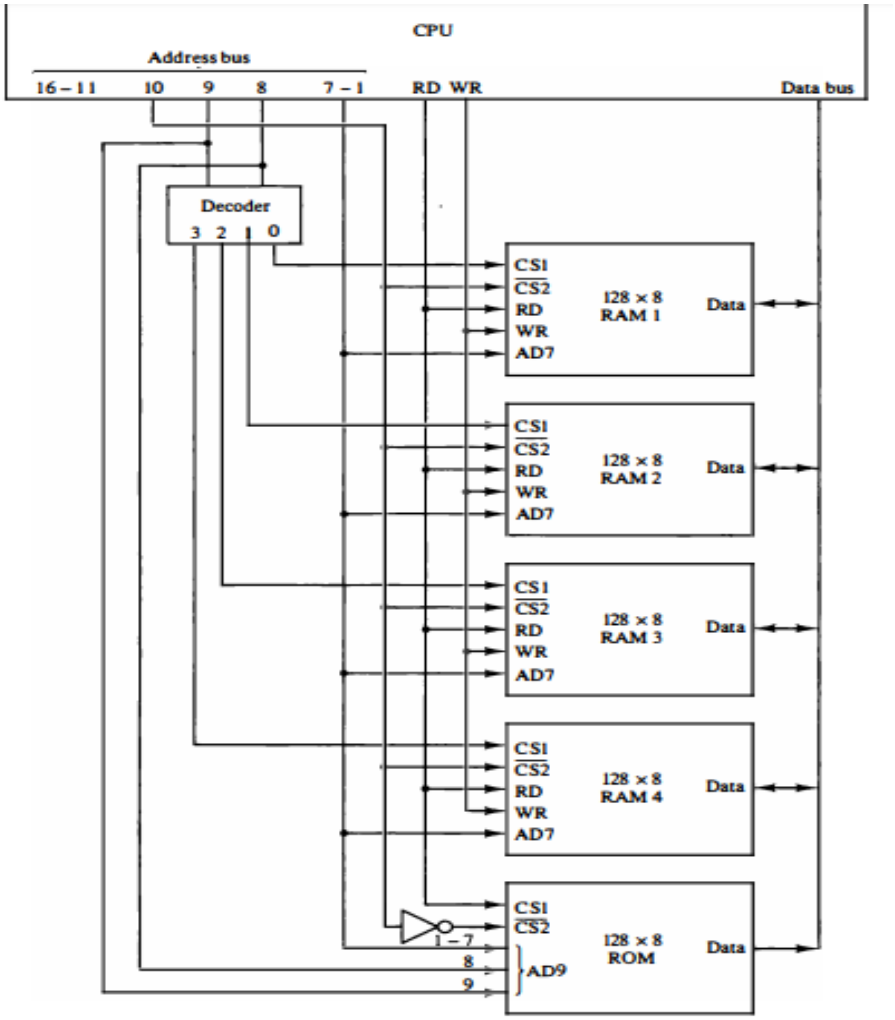
11. A virtual memory system has an address space of 8k words, memory space of 4k words and Page & Block size of 1k words. The following page reference changes occur during a given time interval. 4, 2, 0, 1, 2, 6, 1, 4, 0, 1, 0, 2, 3, 5, 7

Determine the four pages that are resident in main memory after each Page reference change if the replacement algorithm used is

(i) FIFO

(ii) LRU

12. Find the address range (In Hexadecimal) for typical RAM and ROM chip of following memory connection to CPU.



13. Write the sequence of micro-operations a computer system performs to execute the register-reference instruction CIR (circular shift right) & INC (increment to AC) and a memory reference instruction ADD to AC.

14. Explain the DMA operation with suitable diagram.

15. The initial content of PC is 7FF. The content of memory at address 7FF is 9A9F. The content of memory at address A9F is 0C35. The content of memory at address C35 is FFFF. List the contents in hexadecimal of registers: Program counter (PC), Address Register (AR), Data Register (DR), and Accumulator (AC) of the basic computer when an ADD instruction is fetched from memory and executed.

16. A computer system executes an instruction stored in its main memory. The Program Counter (PC) contains the address of the next instruction to be executed. When the CPU begins processing, it fetches the instruction from memory, decodes it to determine the

required operation, and then executes it using the Arithmetic Logic Unit (ALU) and relevant registers. Draw and explain the flowchart for Instruction Cycle of the CPU, showing clearly the register-reference, memory-reference and I/O-reference.

17. The initial content of the Accumulator (AC) is 8C70, and the E flip-flop contains 1.
A CLE (Clear E) instruction is followed by a CIR (Circular shift right through E) instruction.
List the final contents of AC and E.

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Assignment 2.3 (Module II)

1. A 4-stage pipeline has stage delays of 150, 120, 160, and 140 nanoseconds, respectively, with 5 ns registers. Calculate the total time to process 1000 data items. Also find the maximum speed-up ratio?

2. The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 nanoseconds. The first stage is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 nanoseconds. The throughput increase of the pipeline is _____%.

3. Explain Parallel Processing. Draw pipeline processing for instruction: (Ai*Bi)+(Ci-Di) for I = 1, 2, 3. Identify the sub-operations in each pipeline stage using registers R1, R2, R3, etc., and present a space-time diagram for a 5-segment pipeline processing 9 tasks.

4. Consider a pipeline having 4 phases with duration 40, 60, 100 and 80 ns. Given latch delay is 20 ns. Calculate (a) Pipeline cycle time (b) Non-pipeline execution time (c) Speed up ratio (d) Pipeline time for 100 tasks (e) Sequential time for 100 tasks (f) Throughput.

5. Illustrate the behaviour of a pipeline using space-time diagram. A non-pipeline system takes 70ns to process a task. The same task can be processed in a eight – segment pipeline with a clock of 20 ns. Determine speedup ratio of the pipeline for 250 tasks?

6. What is an instruction cycle and write the phases of Instruction cycle used in the computer system for the complete execution of the memory reference, register reference and I/O reference instructions? With a neat flowchart, explain how the timing and control unit determines the instruction after decoding an instruction.

7. Draw and explain Timing and Control unit of Basic computer system and explain the flowchart for instruction cycle?

8. Write the sequence of micro-operations required to perform the memory-reference instructions (i) LDA (Load to AC), (ii) SUB from AC and register-reference instructions (i) SHR (Shift Right), (ii) CMA (Complement AC).

9. The initial content of PC is 5A0H. The instruction stored at 5A0H is 87B6H. The memory content at address B6H is 2E4H. The memory at address E4H contains FFFFH. Show the content of PC, AR, DR, and AC after an AND instruction is fetched and executed if the content of AC initially is 9012H.

10. A computer employs RAM chips of 256x8 and ROM chips of 1024x8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units each with four registers. A memory-mapped I/O configuration used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.

a. How many RAM and ROM chips are needed.
b. Draw a memory-address map for the system.
c. Give the address range in hexadecimal for RAM, ROM and interface.

11. Discuss the Memory Hierarchy in computer system with regard to Speed, Size and Cost? Explain about main memory and its types.

12. A two-way set associative cache memory uses blocks of four words. The cache can accommodate a total of 2048 words from main memory. The main memory size is 128Kx32. Formulate all pertinent information required to construct the cache memory. What is the size of cache memory.

13. A virtual memory system has an address space of 8K words, a memory space of 4K words and page & block sizes are of 1K words. The following page is referenced in memory: 4 2 0 1 2 6 1 4 0 1 0 2 3 5 7. Determine the four pages that are resident in main memory after each page reference change if page replacement algorithm is: FIFO, LRU, Optimal.

14. What is virtual memory? Explain the relation between address space and memory space in a virtual memory system along with its memory table for mapping? Explain Virtual address Mapping using Pages with necessary examples.

15. Find the page faults when the following pages are requested to be loaded in a page frame of size 3: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 using
1) FIFO
2) LRU
3) Optimal Page Replacement
Find the number of misses and hits while using the optimization.

16. Elaborate how DMA bypasses CPU and speeds up the memory operation? With a neat schematic, Explain about DMA controller and its mode of data transfer.

17. What is the need of I/O interface and explain an example I/O interface with neat diagram.

18. Why does a DMA controller use bidirectional read/write control lines? Explain the role of these lines with a neat block diagram of DMA operation and discuss when they function as inputs and outputs.

19. The initial content of the Accumulator (AC) is 6A3F, and the E (carry) flip-flop contains 1.

A CMA (Complement Accumulator) instruction is to be executed, followed by a CIL (Circular shift left through E) instruction. List the contents of the Accumulator (AC) (in Hex) and E flip-flop after each instruction is executed. Also, state the function performed by each instruction.